

Wireless Sound

High end home theatre systems now support wireless streaming for surround sound speakers, thus cutting down on cable clutter

Further reading

For more on home theatre set-ups, read the March 2009 issue of Digit

Place the front left and right speakers at an inward angle. Ensure that they are both equidistant on both sides of the TV and from the listening area. They should be placed at ear level for maximum effect and should form an approximately 45 degree angle from the seating location. The left and right surround speakers need to be placed on both sides of the seating location or slightly behind it. They should be placed approximately 2 feet above ear level for the proper effect.

7.1 SETUP

Most DVDs use 5.1 surround sound while newer high definition formats like Blu-ray contain 7.1 sound tracks. 7.1 systems have two extra surround speakers to give the perception of depth to the home theatre. However, these are useful only if you have a large room or at least more than 3-4 feet of space between your seating location and the back wall. The rear surround speakers should be placed directly behind the listening area at the same height as the side surround speakers, at approximately a 60 degree angle to the listening area.

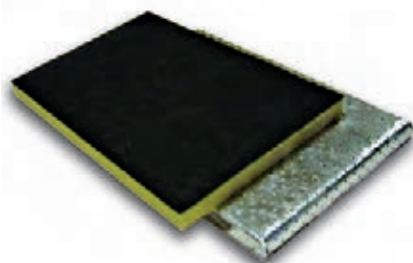
Wiring

The cables and wiring of the home theatre system are also a neglected part of the whole setup. While cheap wiring is of course a no-no, it also doesn't make sense to buy some of the exotic and extremely expensive cables/connectors available in the market. A cable's job is to make sure the signal is transmitted.

A gold plated cable over a normal good quality one will not make a difference in this case. The longer the length of the cable, the smaller 'gauge' cable you need. Normal 12 AWG oxygen free multi-strand cables are good enough for anything till 200 feet. Any audio speciality store will stock the required cables. Just don't fall for the extremely expensive pre-packaged *quality* cables that claim to do everything including improvements in sound quality.

More importantly, power and audio/video cables must be separated at the

back of the home theatre system. A jumbled mess here can result in electrical interference and less than stellar sound quality from the speakers. Avoid cabinets or racks with one or two holes to run cables through them.



Absorbent materials reduce noise

These tend to cause interference since they combine power and audio/video signals. The metal edges may also tend to cut the cables if pulled. Ensure all connectors have good contact and are tightly coupled. Lighting dimmers as well as loose power plugs can also result in hum.

General dos and don'ts

Double check all speaker wires to ensure that the wires are connected to the correct polarity, positive to positive and negative to negative. While

it sounds minor, a mistake here can result in mismatched sounds from the speakers while you go crazy with the settings trying to figure out what went wrong.

Power down all equipment while making the initial connections. Switch on the system only after all connections have been securely made.

Good quality audio equipment including audio receivers and speakers are often quite heavy. Make sure that the cabinet/shelves/speaker stands used can take the load. You don't want any of that expensive stuff falling down after a couple of days.

Avoid placing sub-woofers and speakers in general inside cabinets unless they come prebuilt. Optimal placement may require you to move them around, something that is not possible when placed in a cabinet. The power of the sub-woofer may also be reduced. In case you do need to place it inside a cabinet, make sure that it's properly braced and padded to avoid unnecessary vibrations.

Cooling is an often ignored part of home theatre systems even though high-end equipment does tend to generate a lot of heat. While passive cooling will be adequate in most cases, semiconductors are designed to operate in a particular temperature range, and proper placement can ensure longer life for your hardware.

Consider additional cooling if the equipment is being placed in a cabinet without ventilation outlets. Heat generating equipment like power amplifiers found in audio/video equipment should be placed on top so that warm air can rise upwards and be exhausted quickly.

Most home theatre systems have integrated setup menus and some kind of sample CDs to calibrate the audio and make sure that surround sound is working properly. If you are investing in high end equipment, it may be a good idea to buy a Sound Pressure Level (SPL) meter to help with proper calibration of the complete setup.

Home theatres, speaker setup or even room acoustics are vast topics that have entire books and many web sites devoted to them. We hope this article gives you an idea of what to expect while setting up your own home theatre system to get the best out of the system. Feel free to experiment with room interiors and speaker placement till you find a setup that is just right for you. ■



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Join the Digit Forum at www.thinkdigit.com/forum/, and discuss this article. You can find others who are also interested in setting up home theatre systems. If you haven't already, the next step in spicing up a home theatre setup is to go the DTH route, but hang on, don't miss our comparison of DTH providers next month (January 2010 issue)